

1.0 Overview of the Project

Universiti Teknologi Malaysia, as one of the best technological universities in the world, has many students who move around every day from their residences to the various faculties. Therefore, efficient campus transportation is no longer just a convenience but a vital necessity. The university operates an internal shuttle bus service running different routes within the university campus connecting various residential colleges, faculties, and administration blocks.

However, the current bus service is not satisfactory to the students due to several reasons such as the lack of active buses with high student density. These issues lead to the inconvenience when using the bus service provided and may affect the daily study plan such as students being unable to attend the class on time due to the delay of the buses.

Thus, this project is to design and develop a new UTM Transport App to ensure that students can get transportation easily on campus. This mobile application focuses on bringing the physical and virtual sides of campus life together through the use of real-time data analysis and mobile applications to maximize efficiency and engagement within the campus ecosystem.

In this project, we will collect the students' suggestions or even complaints, and analyse the pain points of the existing transport system. After that, the existing features in the current system will be modified and the new features will be created based on students' requirements. All of these features will be included in our UTM Transport App. Therefore, instead of switching various mobile applications, students can get satisfactory transportation by using only a single mobile application.

2.0 Problem Statement

At a campus as huge as Universiti Teknologi Malaysia (UTM), transportation is not just a "nice-to-have" feature, it is the heartbeat of our daily academic lives. However, the current transportation system tends to be fragmented and traditional that just cannot keep up with the intense schedule of a modern student. For most of us, arranging our traveling schedule is not the smooth transition it should be. It has become a daily source of stress and friction that drains our focus and productivity. To alleviate this issue, we need to address five specific "pain points" that are currently making life harder for the UTM community:

1. The Bus Arrival Uncertainty

Right now, the biggest headache is the total lack of transparency. We are forced to stand at bus stops for who-knows-how-long, often sweating under the scorching sun or getting soaked in a heavy rain, without any idea if the bus is two minutes away or twenty. This uncertainty causes a domino effect, such that we end up late for crucial lectures, miss the start of lab sessions, and live in a constant state of "transit anxiety" that makes it impossible to plan our study groups or even a simple lunch break effectively.

2. The Peak-Hour Seat Shortages

During the morning and evening rush, the campus transport system basically turns into "survival of the fittest" mode. This is because we have no way to check how full a bus is in advance, we often have to watch bus after bus drive past because they are packed with people from the previous stations. It is not just about the delay, but also about the discomfort of being squeezed into overcrowded vehicles and the safety risks that come with it, leaving those at later stops completely stranded.

3. The Inefficient Carpooling

So many of us are willing to share rides or lend a helping hand to ease the load on the buses, but there is not a safe, official way to make that happen. We are currently stuck using disorganized social media chats or word-of-mouth, which is awkward and unreliable. This "fragmented carpooling" means thousands of seats go empty every day while other students are left waiting at the station or walking with their bare feet, wasting a huge opportunity to build a better campus community and lower our carbon footprint.

4. No Route Planning Tools

UTM is huge, and for "freshies" or visitors, it can feel like a total maze. Without a smart tool to help plan a route, students struggle to find the fastest way to get from their residential colleges to distant faculties or admin buildings. We end up wasting time taking long, inefficient paths or walking much further than we need to, simply because we do not have a map that shows us how to navigate the transit transfers correctly.

5. The Lack of Feedback

The most frustrating part of all this is that when things go wrong, we have no real voice. There is not a dedicated, systematic way to report things like a bus being consistently late, a vehicle being dirty, or safety concerns with driving. Without this feedback, the university admin is essentially "flying blind," making it impossible for them to see where the service is failing or how to fix it. This silence leads to a cycle of the same problems happening semester after semester with no end in sight.

3.0 Proposed Solutions

- **Design a Secure User Management Module**

This module focuses on account management functions such as account creation, authentication, and user-profile maintenance to ensure that users of the application are indeed verified UTM students. They need to sign up using UTM email. So, users just need to register their account and click on the "Remember Me" checkbox if one wants quick future logins.

- **Implement a Live Tracking & Mapping Module**

This module offers users the ability to view live information on campus buses' routes and locations. The information displayed consists of the actual location of the bus on the map, along with an Estimated Time of Arrival (ETA) at their current station. The module enables users to pick any bus on the map and observe its movement, along with its plate number.

- **Implement a Seat Availability Monitor Module**

This module allows users to view real-time seat availability of approaching campus buses. The information displayed includes the current occupancy level (available, moderate, or full). So, users can select a bus and check its seat status along with its route, helping them decide whether to board or wait for the next bus.

- **Create a Carpool Module**

This module helps in making travelling on the campus flexible since it offers the students the ability to form rides with other students. Users have two options in this module either to form their own carpool ride or join others. Users can go through the list of available rides and find someone who matches.

- **Design a Route Planning Module**

This module acts as the smart guide in planning how users can travel on the campus in the best way. The fastest route options will be provided together with step by step instructions as well as any required bus transfer along the way. Users can simply input the destination point from the point of departure to access various route options.

- **Implement a Feedback & Rating Module**

This module allows the users to help in improving campus transport through feedback. Here, users can rate their experience riding the bus and report problems such as lateness and overcrowding among others. Once the trip ends, a rating window pops out where users can rate the bus.

6.0 Summary of Requirement Analysis Process

In order to propose an ideal transport system for all UTM students, we involved three information gathering processes which are Joint Application Design (JAD), Questionnaire, and Structured Observation of the Environment (STROBE). From these processes, we noticed that students are highly dependent on the buses on campus. However, the current transport system is not satisfactory to the students and we identified several problems from these three processes.

Firstly, the insufficient real time information of live bus locations and occupancy levels leads to the waste of students' time and the disruption of the plan. Therefore, a Live Tracking & Mapping feature is needed to provide the bus details and the Estimated Times of Arrival (ETA) for all campus buses. Besides that, a Seat Availability Monitoring feature is also required to process the real time occupancy level of each campus bus to ensure students can modify their plan in time if the bus is full.

Next, when students need to go somewhere they are not familiar with, the current system only provides a link to the Google map and there is no guide for students to go to the destination using the free buses provided by UTM. Thus, an Intelligent Route Planning is created to provide a guide and help students in planning the most efficient route by using the free buses to minimize the fee.

Moreover, the students showed dissatisfaction about the lack of official ways to report repeated issues causing the existing issues such as bus lateness to happen repeatedly. In order to reduce those issues, a Feedback and Reporting feature is essential which allows students to rate and submit complaints so that the administration can address those issues effectively.

Furthermore, the current transport system only provides bus services and there is no alternative transport for students to choose. This has led to the creation and use of informal social media groups among students to get transport easily. However, these informal groups are less reliable and may bring risk to both passengers and drivers. Therefore, in order to meet the students' requirements, a Carpool Management is involved in our new transport system to provide another choice for their journey.

There are also several non-functional requirements that make the system operate and ensure the convenience of all UTM members when using it. For instance, the system shall operate automatically from drivers' perspective so that the long-time training sessions for the bus and carpool drivers can be avoided. Besides that, the user interface design focuses on the ease of use to ensure the system can be widely accepted and used by all UTM students as passengers.

From the security perspective, our whole system will require authorization using UTM's official email during the login process. The security level of our transportation can be increased when all the bus drivers, the carpool drivers, and the passengers verify their identity using UTM's official email. Moreover, administration requires unique authentication credentials to monitor network health and view feedback reports. Lastly, our system will also implement strict data encryption to protect the privacy of data such as the user profiles and real-time location data.

In conclusion, we have collected and analysed the problems that occur in the current UTM transport system. In order to solve them, we have analysed the functional and non-functional requirements and proposed the appropriate solutions to increase the satisfaction of users. Furthermore, logical data flow diagrams of the current system were drawn to increase our understanding of the current system and make the upgrading process easier to implement.